

The Biscuit-Piece Pairing Problem

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Abstract

We study a simple biscuit-piece packaging problem: given n distinct biscuit pieces, how many different ways are there to select two different pieces to be placed together? We show that the answer is $\binom{n}{2} = n(n-1)/2$. Two proofs are given. The first counts unordered pairs directly; the second models the pieces as the vertices of a complete graph and counts its edges. We also distinguish the number of pairwise moves from the number of arrangements when the initial, unpaired arrangement is included.

1 Introduction

The phrase “number of packagings” can be ambiguous. In this paper, a *pairwise packaging move* means choosing exactly two distinct biscuit pieces and placing those two pieces together. The order in which the pieces are named does not matter: selecting piece i with piece j is the same move as selecting piece j with piece i .

Definition 1. Let $B = \{b_1, b_2, \dots, b_n\}$ be a set of n distinct biscuit pieces. A *pairwise packaging move* is a two-element subset $\{b_i, b_j\}$ of B .

Thus, the problem is to count the two-element subsets of an n -element set. Standard counting methods for this type of problem are discussed in combinatorics texts such as Bóna [1].

2 By Counting

Theorem 1. For $n \geq 0$ distinct biscuit pieces, the number of pairwise packaging moves is

$$\binom{n}{2} = \frac{n(n-1)}{2}.$$

Proof. There are n choices for a first biscuit piece. Once it has been chosen, there are $n - 1$ choices for a different second piece. This gives $n(n - 1)$ *ordered* selections.

However, each pair has been counted exactly twice. For example, the selection “first b_1 , then b_2 ” and the selection “first b_2 , then b_1 ” both describe the same unordered pair $\{b_1, b_2\}$. Therefore, dividing by 2 gives the number of distinct pairwise packaging moves:

$$\frac{n(n - 1)}{2} = \binom{n}{2}.$$

□

For example, the numbers of pairwise moves for the first few values of n are shown below.

Number of pieces n	0	1	2	3	4	5
Pairwise moves $\binom{n}{2}$	0	0	1	3	6	10

3 Using Graph Theory

The same result has a natural graph-theoretic interpretation. Basic graph concepts and counting techniques also appear in Kreyszig [2].

Definition 2. *The complete graph K_n is the simple graph with n vertices in which every two distinct vertices are joined by exactly one edge.*

Represent biscuit piece b_i by a vertex v_i . Draw an edge $v_i v_j$ exactly when pieces b_i and b_j form a permitted pairwise packaging move. Since every two distinct pieces may be paired, the resulting graph is K_n .

Figure 2 shows the graph interpretation. The first diagram is a path on three vertices; it has only two edges and therefore does *not* represent all possible pairs of three pieces. The remaining diagrams are the complete graphs K_3 , K_4 , and K_5 . Their edges represent all possible pairwise packaging moves.

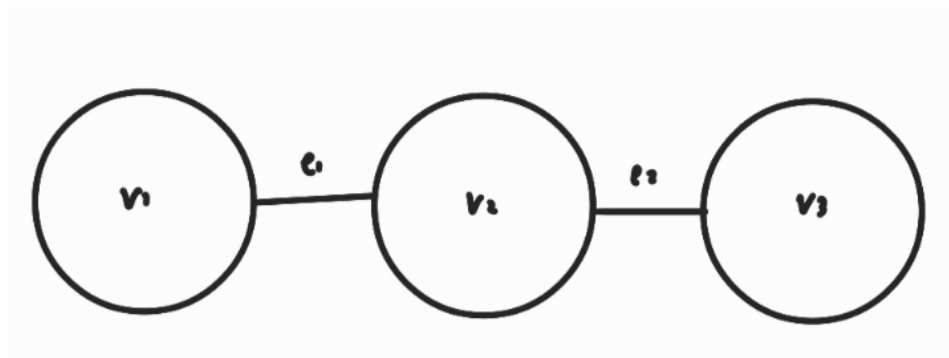


Figure 1: A path on three vertices. It omits the pair $\{v_1, v_3\}$, so it is not the correct graph for the pairing problem.

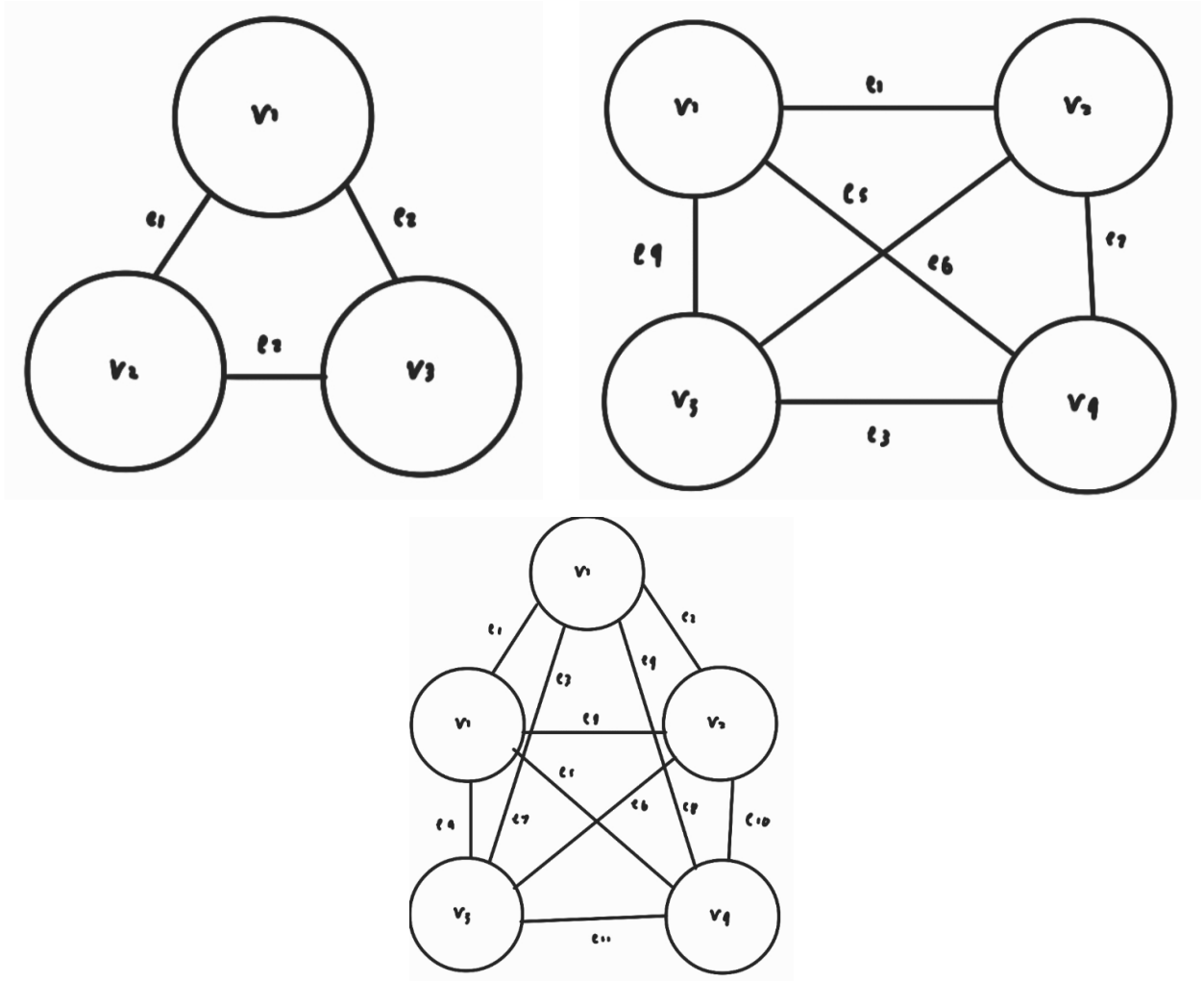


Figure 2: The complete graphs K_3 , K_4 , and K_5 , with respectively 3, 6, and 10 edges.

Theorem 2. *The graph K_n has $\binom{n}{2}$ edges. Consequently, the number of pairwise packaging moves for n biscuit pieces is $\binom{n}{2}$.*

Proof. An edge of K_n is determined by choosing its two distinct endpoints. Thus, the map

$$\{b_i, b_j\} \mapsto v_i v_j$$

is a bijection from the set of pairwise packaging moves to the edge set $E(K_n)$. The number of edges is therefore the number of two-element subsets of an n -element set, namely $\binom{n}{2}$. By the counting argument above, this equals $n(n-1)/2$. $\square$

4 Initial Arrangement

The formula $\binom{n}{2}$ counts only pairwise packaging moves. If the original arrangement in which no pieces have been paired is also called a “packaging,” then it is an additional case. The total number of counted arrangements is then

$$1 + \binom{n}{2} = 1 + \frac{n(n-1)}{2}.$$

For instance, this gives 4 arrangements for three pieces, 7 for four pieces, and 11 for five pieces. Keeping these two quantities separate avoids the ambiguity in the original statement of the problem.

References

- [1] Miklós Bóna, *A Walk Through Combinatorics*, 4th ed., World Scientific, 2017.
- [2] Erwin Kreyszig, *Advanced Engineering Mathematics*, 10th ed., John Wiley & Sons, 2011.